

535/1
PHYSICS
Paper 1
(Theory)
Oct. /Nov. 2026
2½ hours



UGANDA NATIONAL EXAMINATIONS BOARD
Uganda Certificate of Education

PHYSICS

Paper 1
(Theory)

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

*This examination paper consists of **two** sections: **A** and **B**.*

*It has **seven** examination items.*

*Section **A** has **three compulsory** items.*

*Section **B** has **two** parts; **I** and **II**. Respond to only **one** item from **each** part.*

*Respond to **five** items in all.*

Any additional item(s) responded to will not be scored.

***All** responses **must** be written in the answer booklet(s) provided.*

Responses to each item should start on a fresh page.

Indicate the item number you are responding to at the top of the page.

Graph paper is provided.

Mathematical tables and silent non-programmable calculators may be used.

SECTION A

Answer **all** the items in this section.

Item 1.

A town council has installed a tall communication tower to improve communication within a rapidly growing trading center. The tower is fitted with a searchlight, a warning light and several security cameras.

During an inspection, Physics learners are asked to determine whether the tower exceeds the minimum height requiring a warning light. They use a pinhole camera placed 80 m from the base. The pinhole camera is 20 cm long and produces an image of the tower 5 cm high.

To verify their result, the learners produce a sharp sound at the base of the tower and hear its echo from the top 0.50 s later.

Support information:

Speed of sound = 340 m/s

Task

As a student of physics;

- (a) Explain the principle of image formation in a pinhole camera and determine the height of the tower using the camera measurements.
- (b) Explain how the echo method can be used to determine the height of the tower and calculate the height using the given data.
- (c) Compare the two values obtained and advise the council whether the tower should be fitted with a warning light if the required minimum height is 20 m.
- (d) Explain why a concave mirror can be used in a searchlight and state what effect changing the position of the light source has on the reflected beam.

Item 2.

A health center in a rural district receives a radioactive source for use in treating patients suffering from certain diseases. The health workers store the source in a specially constructed concrete room. Some community members become worried after seeing warning signs around the room and wonder why such a small quantity of material requires such strict precautions.

During an inspection, the radiation intensity measured outside the storage room is found to depend on the thickness of the concrete barrier as shown below in **table 1**.

Table 1

Thickness of concrete (cm)	0	10	20	30	40
Radiation intensity (counts per minute)	400	120	50	15	5

The health center considers an intensity of more than 75 units unsafe.

A patient later undergoes a medical examination in which highly penetrating radiation is used to obtain an image of a suspected bone fracture.

Task:

As a student of physics;

- (a) Determine the minimum thickness of concrete that should be used and explain why the radioactive source requires a specially constructed room.
- (b) Explain the difference between irradiation and contamination and state two precautions the health workers should observe when handling the source.
- (c) Explain how radioactive materials can be useful in the treatment and diagnosis of diseases.
- (d) Explain how the radiation used to obtain the image of the patient's bone fracture is able to produce the image and state one other application of the radiation.

Item 3.

A group of Senior Four learners visits a coastal area during a school study tour. Early in the morning they notice that the sea level is unusually low. Several hours later, the water has risen and covered part of the beach.

That evening, the learners observe the Moon and notice that its appearance is different from what they saw several nights earlier. Their teacher explains that the same principles of Earth and space physics help to explain tides, the phases of the Moon and the changing seasons experienced in different parts of the world.

The learners also learn that fishermen in the area use GPS equipment to locate their boats when visibility is poor.

Task:

As a physics learner;

- (a) Explain the cause of the changing water level observed at the coast and state two ways in which tides are useful to people.
- (b) Explain why the Moon appears to change its shape when observed from Earth and distinguish between a natural and an artificial satellite.

(c) A learner claims that countries experience different seasons because they are at different distances from the Sun. Explain why this statement is not correct and give the actual cause of seasons.

(d) Explain how the GPS system can help the fishermen locate their boats and state two other applications of artificial satellites.

SECTION B

Part I

Answer **one** item from this part.

Item 4.

A group of students visits a farm where the owner is trying to improve the efficiency of his machines. A loaded cart is pushed up a ramp into a store. The owner notices that pushing the cart up the ramp requires less effort than lifting it directly, although the distance moved is greater.

At the same farm, a water tank is supported above the ground. The farmer wants to know how much pressure the water exerts on the bottom of the tank. He also wants to install a hydraulic press for compressing agricultural waste.

Support information:

Mass of cart = 120 kg

Height of store = 2 m

Length of ramp = 8 m

Acceleration due to gravity = 10 m/s²

Task:

As a physics learner;

(a) Explain why the inclined plane makes it easier to raise the cart and calculate the minimum ideal effort required to move it to the store.

(b) Explain why the actual effort will be greater than the ideal value and state two factors responsible for the difference.

(c) The water tank is 5 m deep. Calculate the pressure due to the water at the bottom and explain why the tank wall should be stronger near the bottom. Take density of water as 1000 kg/m³.

(d) Explain the principle of operation of the hydraulic press and give two reasons why it is useful in agricultural work.

Item 5.

A school laboratory is investigating why some materials are better suited for particular uses. During the investigation, students compare a metal saucepan, a wooden handle, a thermos flask and a metal bridge.

They heat a quantity of water and observe that the metal saucepan becomes hot quickly while the wooden handle remains relatively cool. They also learn that bridges are constructed with gaps and that thermos flasks are designed to reduce heat transfer.

Task:

As a physics learner;

- (a) Explain the suitability of the materials used for the saucepan and its handle in terms of heat transfer.
- (b) A 2 kg quantity of water is heated from 25 °C to 85 °C. If the specific heat capacity of water is 4200 J/kg, determine the heat gained by the water.
- (c) Explain using a relevant diagram how a thermos flask reduces heat transfer by conduction, convection and radiation.
- (d) Explain why gaps are provided in bridges and railway lines and state one problem that may occur if no allowance is made for expansion.

Part I1

Answer **one** item from this part.

Item 6.

A businessman operates a laundry using two electric irons, a washing machine and several lamps. He receives a monthly electricity bill that is much higher than expected and believes that the electricity company has overcharged him.

His appliances are:

Appliance	Rating	Daily use
Two irons	1200 W each	2 h
Washing machine	2200 W	3 h
Three lamps	100 W each	10 h

Electricity costs UGX 750 per kWh.

The businessman also wants to understand the motor inside the washing machine and ways of reducing his bill.

Task:

As a learner of physics;

- (a) Determine the electrical energy consumed by the appliances in one day and the expected cost per day.
- (b) Determine the expected cost for 30 days and advise the businessman whether his bill is reasonable, ignoring any fixed charges.
- (c) Explain how the electric motor in the washing machine produces rotation and state two factors that affect its turning effect.
- (d) Advise the businessman on three practical ways of reducing electrical energy consumption without stopping his business operations.

Item 7.

A small hydroelectric project is being planned to supply electricity to a rural trading center. The engineers intend to generate 50,000 W of alternating current at 500 V and transmit it over a distance of 10 km.

The total resistance of the transmission cables is $4\ \Omega$ per kilometer. However, appliances in homes require a supply of approximately 240 V.

The community members want to understand why electricity is generated as AC, why its voltage is changed before transmission and how electrical losses can be reduced.

Task;

As an aspiring physicist;

- (a) Explain how an AC generator produces electrical energy at the hydroelectric station and state the energy transformations involved.
- (b) Determine the current supplied by the generator and the power lost in the transmission cables if electricity were transmitted directly at 500 V.
- (c) Explain how transformers can be used to reduce power loss during transmission and why a step-down transformer is required before electricity reaches homes.
- (d) State three precautions or protective devices that should be used in domestic electrical installations and explain the importance of any two.